

-- Cox Regression - Sarcopenia Onset -----

i Definition : fnih
i Covariates : time_dependent
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 2 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 3 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 6 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 7 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1950 intervals | 34 events | 7795.8 person-years
Fitting .imp = 8 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 9 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1950 intervals | 34 events | 7795.8 person-years
Fitting .imp = 10 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

??? Time-dependent dataset: 1951 intervals | 33 events | 7798.4 person-years
Fitting .imp = 12 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 13 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 14 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 15 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1951 intervals | 33 events | 7798.4 person-years
Fitting .imp = 16 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

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Fitting .imp = 18 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants
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-- Time-dependent covariates dataset (one row per pt x visit) --
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??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 19 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants
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-- Time-dependent covariates dataset (one row per pt x visit) --
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??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 20 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants
```

```
-- Time-dependent covariates dataset (one row per pt x visit) --
```

```
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
```

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-- Pooled adjusted model results (Rubin's rules) --
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# A tibble: 8 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>
1 dairy_exposure 0.999 0.997 1.00 0.667 2 BMI 1.17 1.08 1.26 0.000251 3
education_levelMedium (ISCED 3-4) 0.621 0.286 1.35 0.216 4 education_levelHigh
(ISCED 5-8) 0.0000000104 0 Inf 0.997 5 smoking_statusFormer 1.05 0.490 2.25
0.894 6 smoking_statusCurrent 0.461 0.0537 3.96 0.461 7 mvpa_min_day_f1 1.000
0.996 1.00 0.917 8 diabetes_statusDiabetes 0.772 0.240 2.49 0.651
```

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

'geom_smooth()' using formula = 'y ~ x'

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Linearity plots saved to

03_outputs/Cox/mice_fnih_time_dependent_continuous_27072026_1812

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 0 observation(s) with |deviance residual| > 3.

??? No extreme deviance residuals (all |dev| <= 3).

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 22 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

Cox-Snell plot saved to

03_outputs/Cox/mice_fnih_time_dependent_continuous_27072026_1812

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

GVIF Df GVIF^(1/(2*Df)) dairy_exposure 1.011 1 1.005 BMI 1.086 1 1.042

education_level 1.010 2 1.002 smoking_status 1.032 2 1.008 mvpa_min_day_f1

1.020 1 1.010 diabetes_status 1.083 1 1.041

??? No collinearity concerns (all VIF/GVIF within range).

-- Adjusted scenario survival curves (time-dependent) [mice_imp1]

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high
1 Q1 (low)	280	7.3	1958.90	3.73	1.53	7.57
2 Q2	279	8.0	1961.70	4.08	1.76	8.04
3 Q3	279	9.3	1936.32	4.80	2.23	9.03
4 Q4 (high)	279	8.0	1940.70	4.12	1.78	8.12

Incidence table saved to

03_outputs/Cox/mice_fnih_time_dependent_continuous_27072026_1812

Model results saved to

03_outputs/Cox/mice_fnih_time_dependent_continuous_27072026_1812/mice_model_results.csv